

How to compile:
g++ -Wall hw5.cpp

How to Run:
./a.out input#.txt FCFS
./a.out input#.txt SJF
./a.out input#.txt RR 5

Test Case 1:

```
Time 0: Process 0 has started.
Time 12: Process 0 has completed.
Time 12: Process 1 has started.
Time 16: Process 1 has completed.
Time 16: Process 2 has started.
Time 17: Process 2 has completed.
Time 17: Process 3 has started.
Time 19: Process 3 has completed.
PID: 0
Arrival Time: 0
Start Time: 0
End Time: 12
Running Time: 12
Waiting Time: 0

PID: 1
Arrival Time: 2
Start Time: 12
End Time: 16
Running Time: 4
Waiting Time: 10

PID: 2
Arrival Time: 3
Start Time: 16
End Time: 17
Running Time: 1
Waiting Time: 13

PID: 3
Arrival Time: 4
Start Time: 17
End Time: 19
Running Time: 2
Waiting Time: 13

Average Waiting Time: 9
```

Test Case 2:

```
Cpu is idle right now  
Cpu is idle right now  
Cpu is idle right now
```

```
0 has started.  
0 has completed.  
1 has started.  
1 has completed.  
2 has started.  
2 has completed.
```

```
PID: 0  
Arrival Time: 3  
Start Time: 3  
End Time: 7  
Running Time: 4  
Waiting Time: 0
```

```
PID: 1  
Arrival Time: 5  
Start Time: 7  
End Time: 9  
Running Time: 2  
Waiting Time: 2
```

```
PID: 2  
Arrival Time: 8  
Start Time: 9  
End Time: 12  
Running Time: 3  
Waiting Time: 1
```

```
Average Waiting Time: 1
```

Test Case 3:

```
Time 0: Process 0 has started.  
Time 3: Process 0 stopped running.  
Time 3: Process 1 has started.  
Time 5: Process 1 stopped running.  
Process 1 has completed.  
Time 5: Process 2 has started.  
Time 8: Process 2 stopped running.  
Time 8: Process 3 has started.  
Time 11: Process 3 stopped running.  
Process 3 has completed.  
Time 11: Process 0 has started.  
Time 13: Process 0 stopped running.  
Process 0 has completed.  
Time 13: Process 2 has started.  
Time 16: Process 2 stopped running.  
Time 16: Process 2 has started.  
Time 17: Process 2 stopped running.  
Process 2 has completed.  
PID: 0  
Arrival Time: 0  
End Time: 13  
Running Time: 5  
Waiting Time: 8  
  
PID: 1  
Arrival Time: 0  
End Time: 5  
Running Time: 2  
Waiting Time: 3  
  
PID: 2  
Arrival Time: 0  
End Time: 17  
Running Time: 7  
Waiting Time: 10  
  
PID: 3  
Arrival Time: 1  
End Time: 11  
Running Time: 3  
Waiting Time: 7  
  
Average Waiting Time: 7
```